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Enterprise Network Security Implementation Using Palo Alto, Wazuh, and Sysmon

CSC-438 · NETWORK SECURITY · FINAL PROJECT

As the final project for my Network Security course (CSC-438), I designed, configured, secured, and monitored an enterprise-style network using industry-standard cybersecurity technologies. The project required implementing multiple layers of network defense, including a next-generation firewall, network segmentation, intrusion detection, host-based monitoring, centralized log management, advanced Windows event logging, and host firewall hardening while maintaining secure communication between multiple network segments.

1 Network Infrastructure Configuration

I began by configuring a Palo Alto Networks virtual firewall from scratch. This involved assigning and configuring all required network interfaces, including the management interface, WAN, LAN, and DMZ interfaces. I configured:

- **Management Interface** — Static IP: [192.168.2.2](#)
- **WAN Interface (ethernet1/1)** — DHCP configuration to obtain an external IP address dynamically
- **DMZ Interface (ethernet1/2)** — Static IP: [192.168.1.1/24](#)
- **LAN Interface (ethernet1/3)** — Static IP: [192.168.2.1/24](#)

I also configured Layer 3 interfaces; security zones (LAN, DMZ, WAN); a virtual router; interface assignments; zone assignments; and static routing where required. This provided complete Layer 3 connectivity throughout the environment.

2 DHCP Configuration

I configured the Palo Alto firewall to function as the DHCP server for the LAN. This included configuring the DHCP address pool, default gateway, DNS server, and DHCP lease configuration. This allowed Windows client systems to automatically receive IP configuration.

3 Outbound NAT Configuration

I implemented Dynamic Source NAT (PAT) so all internal devices could securely access the Internet using the WAN interface. The NAT policy included Source Zone: LAN; Destination Zone: WAN; Source Address: Any; Destination Address: Any; Dynamic IP and Port Translation; and Interface Address Translation. This allowed Internet connectivity without assigning public IP addresses to internal hosts.

4 Palo Alto Security Policy Implementation

I designed a layered firewall policy based on the principle of least privilege. Instead of allowing unrestricted communication, I created individual security rules permitting only required services.

Internet Access Rule

Configured LAN clients to browse the Internet. Allowed HTTP, HTTPS, and DNS, with an Antivirus Security Profile applied.

HTTP Access Rule

Created a dedicated rule allowing users on the LAN to access the Windows Server 2019 web server hosted in the DMZ (Source Zone: LAN, Destination Zone: DMZ, Destination: Windows Server 2019, Service: HTTP Port 80).

FTP Access Rule

Created a custom FTP service object (TCP, Destination Port 21) and a security rule allowing LAN to the Windows Server FTP service.

Wazuh Communication Rule

Created multiple custom service objects required by Wazuh — TCP 1515, UDP 1514, TCP 55000, and TCP 443 / 5601 (Dashboard). I grouped these into a Service Group named `wazuhgrp` and created a dedicated security policy allowing communication between LAN systems and the Wazuh Manager in the DMZ.

DMZ Communication Rules

Created additional firewall policies allowing DMZ → WAN for updates and DMZ → LAN for required Wazuh communication.

5 Antivirus Security Profiles

Applied Palo Alto Antivirus Profiles to all firewall rules permitting user traffic. This enabled inline malware inspection using the Next-Generation Firewall.

6 Malware Prevention Testing

Validated firewall malware protection using the standard EICAR antivirus test file. I performed a download attempt, antivirus inspection, and threat prevention validation, and confirmed the Palo Alto firewall detected and blocked the malicious test file before the download completed.

7 Wazuh Deployment

Installed Wazuh as the Host-Based Intrusion Detection System (HIDS). I configured a CentOS 8 server as the centralized Wazuh Manager and installed Wazuh Agents on:

- Windows Server 2019
- Windows Server 2022
- Windows 11 Client 1
- Windows 11 Client 2

I configured secure communication between agents and manager, performing agent registration, authentication, key exchange, and connectivity verification, and validated agent communication inside the Wazuh dashboard.

8 Wazuh Troubleshooting

Diagnosed multiple deployment issues including firewall communication problems, agent authentication failures, service startup failures, agent registration issues, Windows Service troubleshooting, and port connectivity verification. Tools used: PowerShell `Test-NetConnection` ; Windows Service Control `sc query` and `net start wazuh` ; and the agent authentication utility `agent-auth.exe` .

9 Sysmon Deployment

Installed Microsoft Sysinternals Sysmon on every Windows endpoint, configured using Olaf Hartong's enterprise Sysmon Modular Configuration. I collected advanced Windows telemetry including:

- | | |
|--------------------------|----------------------------|
| • Process Creation | • Registry Changes |
| • Parent/Child Processes | • Network Connections |
| • PowerShell Activity | • Image Loading |
| • Command Line Arguments | • Process Injection Events |
| • Driver Loading | • Hash Generation |
| • DLL Loading | • Event IDs |
| • File Creation | • Security Events |

10 Sysmon and Wazuh Integration

Modified every Windows Wazuh Agent configuration, adding:

```
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

This enabled centralized collection of Sysmon events into Wazuh. I validated agent communication, event forwarding, log ingestion, Sysmon visibility, and endpoint telemetry using the Wazuh Discover dashboard.

11 Windows Defender Firewall Hardening

Configured Windows Defender Firewall on every Windows system.

- **Windows Server 2019** — Allowed HTTP and FTP
- **Windows Server 2022** — Allowed Active Directory Services
- **Windows 11 Clients** — Blocked all inbound connections except existing required system rules

Configured the default policy to block inbound traffic and allow only explicitly permitted services, and configured outbound communication for Internet browsing, internal services, and Wazuh communication.

12 Network Segmentation

Maintained strict separation between LAN, DMZ, and WAN while ensuring required services remained operational. I verified connectivity for LAN → Web Server, LAN → FTP Server, LAN → Wazuh Dashboard, and LAN → Internet, without exposing unnecessary services.

13 Enterprise Security Principles Applied

Throughout the project I implemented multiple security concepts including:

- Defense in Depth
- Network Segmentation
- Principle of Least Privilege
- Next-Generation Firewall Administration
- Network Address Translation (NAT)
- Stateful Firewall Policy Design
- Intrusion Detection
- Host-Based Intrusion Detection
- Centralized Security Monitoring
- Endpoint Logging
- Security Event Correlation
- Malware Prevention
- Threat Detection
- Windows Event Collection

- Security Hardening
- Firewall Rule Optimization

- Windows Endpoint Security
- Enterprise Logging Architecture

14 Technologies Used

- Palo Alto Networks VM-Series Firewall
- Wazuh SIEM/HIDS
- Wazuh Agent
- Wazuh Dashboard
- Sysmon (Microsoft Sysinternals)
- Olaf Hartong Sysmon Modular Configuration
- Windows Server 2019
- Windows Server 2022
- Windows 11
- CentOS 8 Linux
- Windows Defender Firewall
- DHCP
- NAT (PAT)
- TCP/IP Networking
- Layer 3 Routing
- Security Zones
- Antivirus Profiles
- FTP / HTTP
- PowerShell
- Windows Services
- Event Viewer
- XML Configuration

This project strengthened my hands-on experience with enterprise network security by combining firewall administration, network segmentation, intrusion detection, endpoint monitoring, malware prevention, and centralized log analysis into a fully functioning security architecture similar to what is used in production enterprise environments.